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THE NEW MICROFOCUS STATION FOR THE NOTOS BEAMLINE AT THE ALBA SYNCHROTRON*

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Abstract

The NOTOS beamline (BL16) at ALBA synchrotron combines X-ray Absorption Spectroscopy (XAS) and X-ray Diffraction (XRD), operating in the 4.5-30 keV range. Since 2022, it has offered two End Stations (ES): one for metrology and XAS, and another combining XAS and XRD. In this work we present a third Microfocus End Station (µFo-ES) planned to be in operation by the end of 2025. It will provide spot sizes below 10×10 µm² with a flux >$7.3\cdot10^{13}$ ph/s/mm², enabling XAS in fluorescence and transmission. It uses the existing beamline optics with the addition of a pair of Kirkpatrick–Baez (KB) mirrors working under high vacuum. The KB positioning system is based on an in-housed developed design and the mirrors will be elliptically bent using ALBA mirror benders with sub-nanometric resolution. High-precision slits included in the µFo-ES will be placed upstream the KB and used for diagnostics and to ensure a correct beam size and collimation. Furthermore, the µFo-ES will integrate a compact sample environment including an ionization chamber, on-axis imaging system, and a fluorescence detector for variable incident angles and small sample to detector distances. To ensure compatibility with downstream ES and prevent photon flux loss, the µFo-ES has been designed to be fully retractable from the beam path.

INTRODUCTION

ALBA is a synchrotron light source facility located near Barcelona, Spain. It currently operates 13 beamlines covering a broad range of scientific applications, with three additional beamlines in various stages of design or construction. Short-term strategic plans include upgrading the facility to a fourth-generation synchrotron source, along with significant enhancements to existing beamlines to support cutting-edge research.

NOTOS is the result from the confluence between the Metrology and Test Beamline proposed for ALBA phase II and the transfer of ESRF's BM25a, devoted to X-ray Absorption Spectroscopy (XAS) and Powder Diffraction (PD). NOTOS beamline operates between 4.5 and 30 keV using a bending magnet as its source. It is a versatile experimental beamline designed for *in situ* and *operando* studies through the combined use of XAS and X-ray Diffraction (XRD). The current setup consists of two experimental End Stations (ES): a multipurpose ES and a PD-XAS ES. The multipurpose ES offers an open-space setup suitable for metrology applications as well as XAS measurements in both fluorescence and transmission modes. On the other hand, the PD-XAS ES is equipped with a two-circle diffractometer, enabling simultaneous XAS (in fluorescence and transmission geometries) and XRD measurements. Additionally, the beamline includes a fully integrated gas handling system with multiple gas lines, flow and pressure control, for a user-friendly operation.

To overcome the current 100×100 µm² spot size limitation on both current ES. a newly developed Microfocus End Station (µFo-ES), placed upstream the multipurpose ES, will be used to increase experimental possibilities by providing beam spot sizes smaller than 10×10 µm² fwhm. Furthermore, to prevent photon flux loss when using the other ES, the µFo-ES has been designed to be fully retractable from the beam path.

END STATION

The µFo-ES is placed downstream a complex set of already working beam conditioning elements and upstream the multipurpose ES (Fig. 1A). Because of the limited available space for the µFo-ES, most of its components must be compact and work in tight spaces. For this reason, the different components of this ES were designed and assembled in-house, even though commercial products with similar functions may be available in the market. The µFo-ES is composed of three main components, in beam path order are: high precision slits, focusing mirrors and a dedicated sample environment. The µFo-ES forms part of In-CAEM, a strategic project at ALBA that seeks to enable correlative measurements between synchrotron beamlines and advanced electron microscopy techniques. To meet this objective, every component of the sample environment is designed with a high degree of flexibility and adaptability, ensuring seamless integration of diverse types of samples. A general view of the µFo-ES is presented in Fig. 1B.

High Precision Slits

High-precision tungsten slits are primarily used to control the beam size, as well as for alignment procedures and diagnostic purposes. These compact slits, extending only 65 mm in the direction of the beam, are positioned at the entrance of the main vacuum chamber and are actuated by SMARACT piezo-driven stages (Fig. 2A). To aid in beam alignment, the upstream face of the slits is coated with a fluorescent material that emits visible light upon exposure to the X-ray beam, enabling accurate visualization and tracking of the beam position.

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Focusing Mirrors

The beam is focused by a pair of elliptically bent mirrors arranged in a Kirkpatrick–Baez (KB) configuration: a horizontally focusing mirror (HFM) and a vertically focusing mirror (VFM). Both mirrors are bent in the meridional direction using high stability ALBA mirror benders, which offer sub-nanometric resolution and optical stability. This configuration enables correction of wavefront distortions induced by static or dynamic factors, such as long-period figure errors and thermally induced deformations [1].

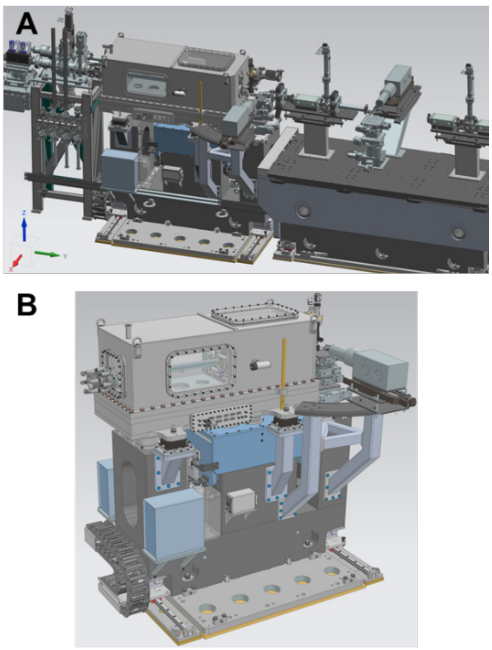

Figure 1: The µFo-ES in NOTOS beam line. A) The µFo-ES is constrained between beam conditioning elements and the multipurpose ES. B) Detailed view of the whole µFo-ES.

The position of the horizontally focusing mirror (HFM) is controlled by a proven flexure-based system operating under vacuum conditions, allowing precise adjustment with resolutions of 0.1 µm in the X direction and 0.05 µrad for the pitch. The positioning system for the vertically focusing mirror (VFM) operates similarly, although it functions under ambient pressure conditions (Fig. 2B). To avoid interfering with experiments conducted at the two downstream ES, both mirrors can be retracted from the beam path, offering a travel range of ±10 mm in the X and Z directions, respectively. However, in this retracted state, the beam still passes through the vacuum chamber and gets absorbed by Kapton windows located at the entry and exit points. To minimize photon flux losses, a retraction system is available to move the entire µFo-ES assembly out of the beam path.

Retraction System

The KB mirror pair is mounted on a granite support to ensure high mechanical stability. During standard operation, the granite rests on a fixed base plate (Fig. 3A). To retract the entire station, two custom-made wedges (Airloc, Switzerland) are manually adjusted to slightly elevate the granite, transferring its load from the base plate to a set of linear guides that enable lateral displacement out of the beam path. Unlike standard wedges, the custom design includes two aligned spherical seats, effectively converting the joint from a spherical to a cylindrical one (Fig. 3B). This design constrains movement in the perpendicular direction and prevents tilting of the granite. To further ensure stability during the transition, additional stabilizing brackets are employed. Furthermore, two mechanical pushers on each side guarantee reproducible positioning, allowing the granite to return to its exact original alignment without requiring further adjustment.

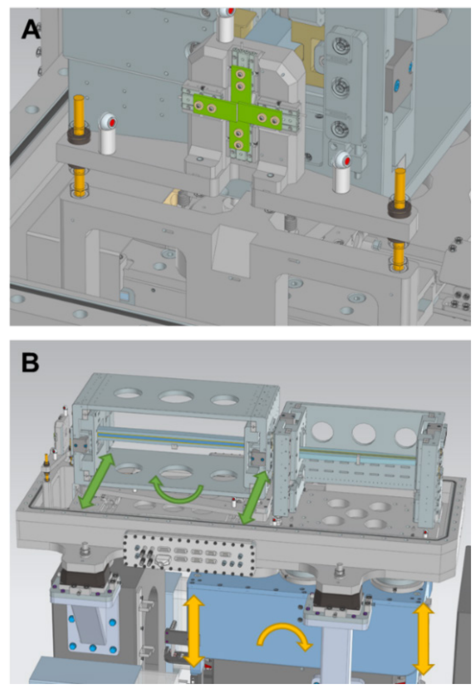

Figure 2: Components inside the vacuum chamber. A) Detailed view of the slits. B) Actuated motions involved in the HFM (green) and VFM (yellow) positioning.

Sample Environment

The focal point of the beam is located 425 mm downstream from the reflection point at the vertically focusing mirror (VFM), leaving only 145 mm outside the vacuum chamber for the installation of all sample instrumentation. This limited space imposes strict constraints on the design, requiring a highly compact and integrated setup.

An MKC12 ionization chamber from ESRF is positioned immediately outside the vacuum chamber to monitor the beam intensity. The chamber is mounted on linear rails, allowing it to be retracted from the beam path when not in use to prevent unnecessary photon loss and to maximize the available flux for experiments.

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To enable precise sample positioning along the beam axis, an on-axis imaging system is installed downstream of the ionization chamber. The system consists of a perforated mirror with a 1 mm central hole (Green-light Solution) placed at a 45° angle relative to the beam and coupled with a Navitar UltraZoom 6000 lens. The lens operates at a working distance of 92 mm and with a resolving power of 4.7 µm. Alignment of the optical axis with the X-ray beam is achieved through a Newport 9064-XYZ translation stage, supplemented by a high-precision rotation stage to fine-tune angular alignment. This setup allows accurate visual monitoring of the sample and beam intersection.

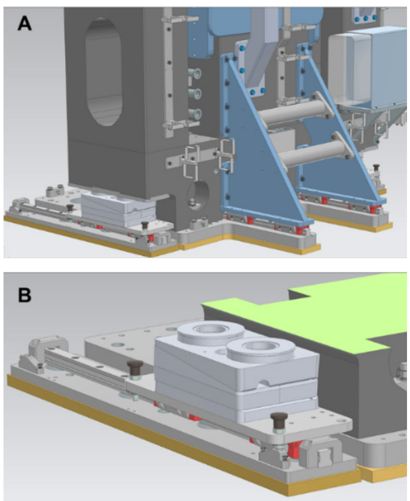

Figure 3: Retraction system. A) Wedges transfer weight from the base plate to lateral linear guides. Dual aligned spherical seats on the wedges (B) and added brackets (A) prevent granite tilting.

Sample positioning is handled by a high-precision stage tower (xHUBER) that provides motorized translation in the XYZ directions and rotational control, with a resolution down to 0.2 µm per step. This enables sub-micrometric accuracy in sample placement, essential for high-resolution measurements.

The sample environment is compatible with the use of two different fluorescence detection systems. The first is a Vortex EX Hitachi Silicon Drift Detector (SDD) (Fig. 4A), equipped with a positioning mechanism that allows adjustment of the incident angle between 40° and 90° relative to the beam. Thanks to its compact design, this detector can operate at close sample-detector distances (approximately 15 mm), which enhances detection efficiency. The second option is a larger fluorescence detector, a Canberra X-PIPS SXD 13 (Fig. 4B), which is already in use at the multipurpose ES at NOTOS beamline. This 13-channel detector provides very fast, low noise response which results in extremely good energy resolution with fast peaking times. As the other detector, it can also work in a range of possible incident angles between 40° and 90° relative to the beam, although the working distance (sample-detector) is slightly larger. This flexibility in detector configurations allows users to tailor the setup to the specific requirements of their experiment.

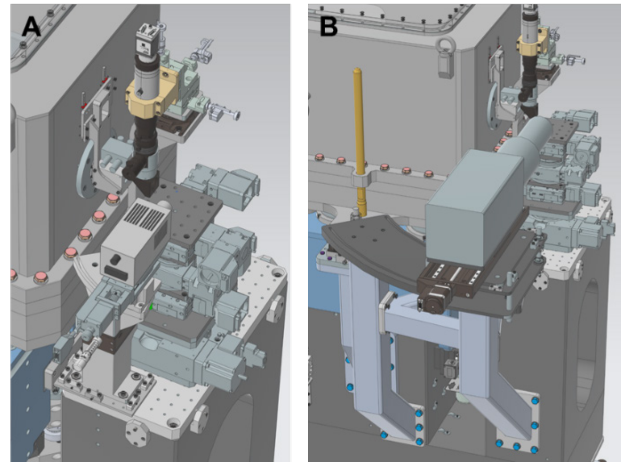

Figure 4: Complete view of the components in the sample environment. With the Vortex EX detector in use (A) and with the Canberra X-PIPS detector (B).

CONCLUSION

In summary, the sample environment at the NOTOS beamline has been carefully engineered to operate within the spatial constraints of the current beam line equipment, offering a compact yet highly functional setup. The use of a KB mirror pair mounted on a stabilized granite support, with a custom retraction system, ensures flexibility without compromising alignment and photon flux loss. Additionally, the system provides a complete sample environment allowing users to tailor the configuration more suitable to their experimental needs. This versatile and modular design supports a wide range of in situ and operando studies while maintaining beam quality and operational efficiency.

The installation of the µFo-ES at the NOTOS beamline is expected to be completed by the end of 2025, with further commissioning and optimization activities scheduled to continue throughout the same year.

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